

Background

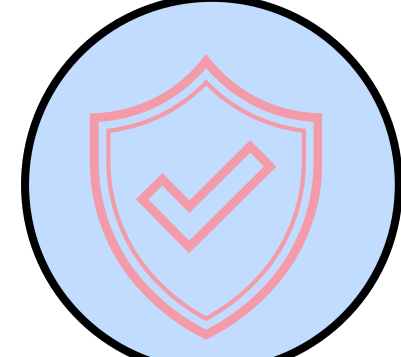
During field trip departure, staff must manage attendance, forms, lunches, supervision, and wristband application all at once. Wristband application becomes a major bottleneck

Opportunity

Redesign the wristband application process to:

- Reduce manual application
- Allow staff to focus on what matters most: supervising children

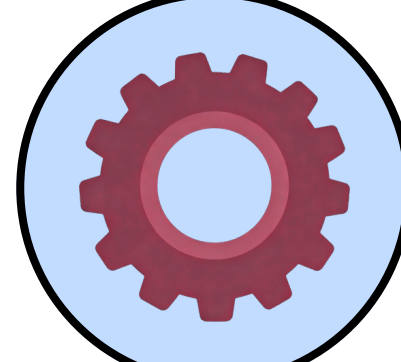
Design Objectives



Efficiency: faster application / lower setup time



Child engagement: low-anxiety, approachable design



Low staff burden: minimal manual intervention



Safety: eliminate sharp hazards and reduce force on wrist

Next Steps

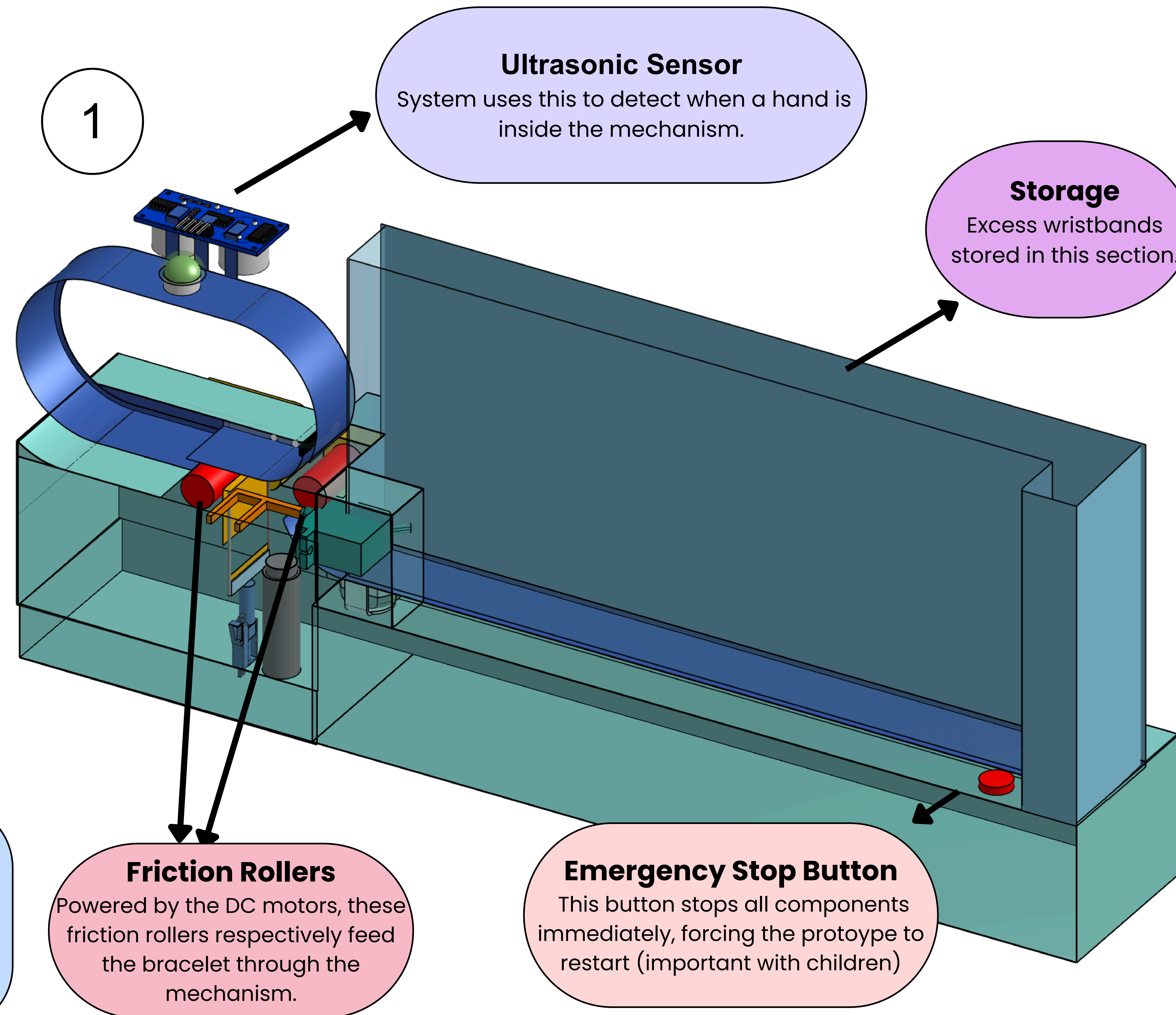
- Integrate full system
- Improve feeding mechanism
- Integrate Excess Management System
- Assess durability (long-term use)
- Select suitable materials
- Upgrade force sensor for adjustable control



The SNAP BUDDY

AUTOMATED WRISTBAND APPLICATION SYSTEM

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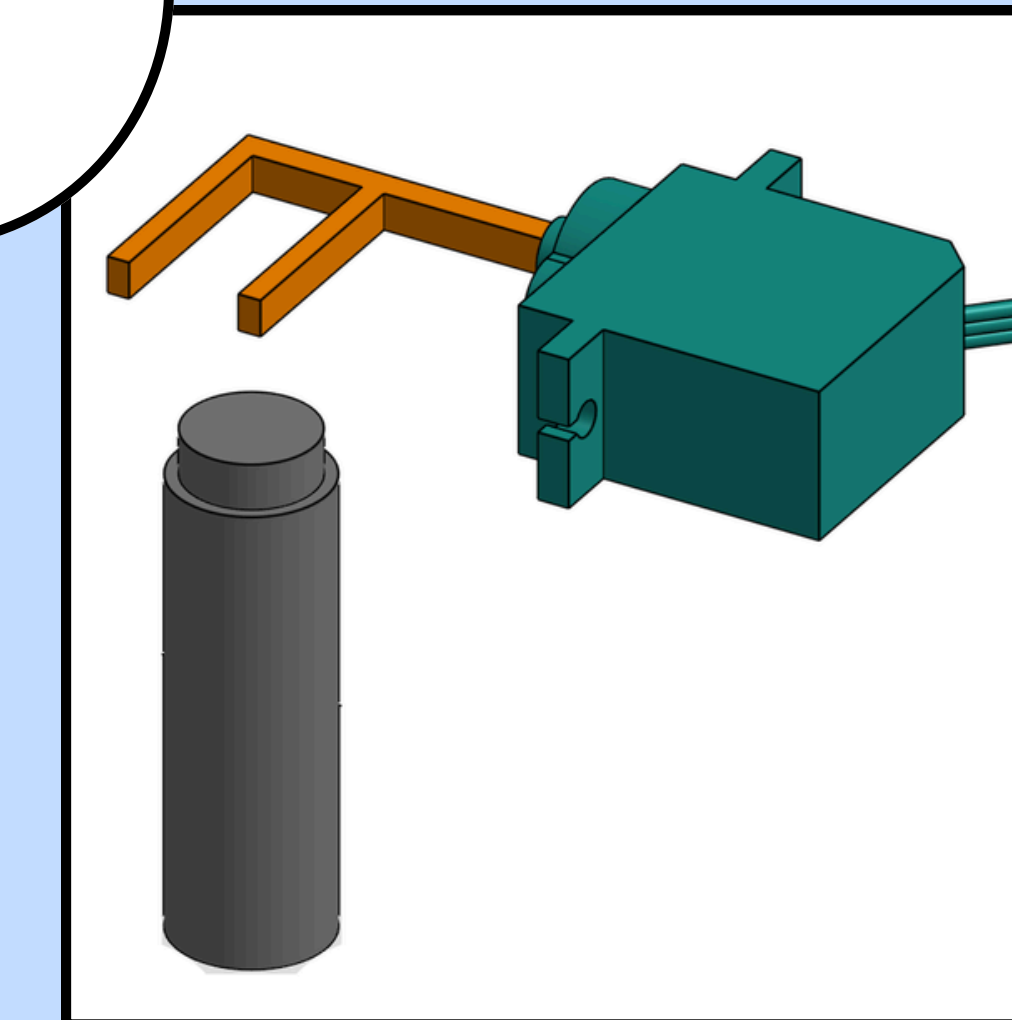
2



Thin Film Pressure Sensor

A sensor whose resistance changes with applied force. The Arduino reads this as a 0–1023 signal, which can be calibrated to estimate force.

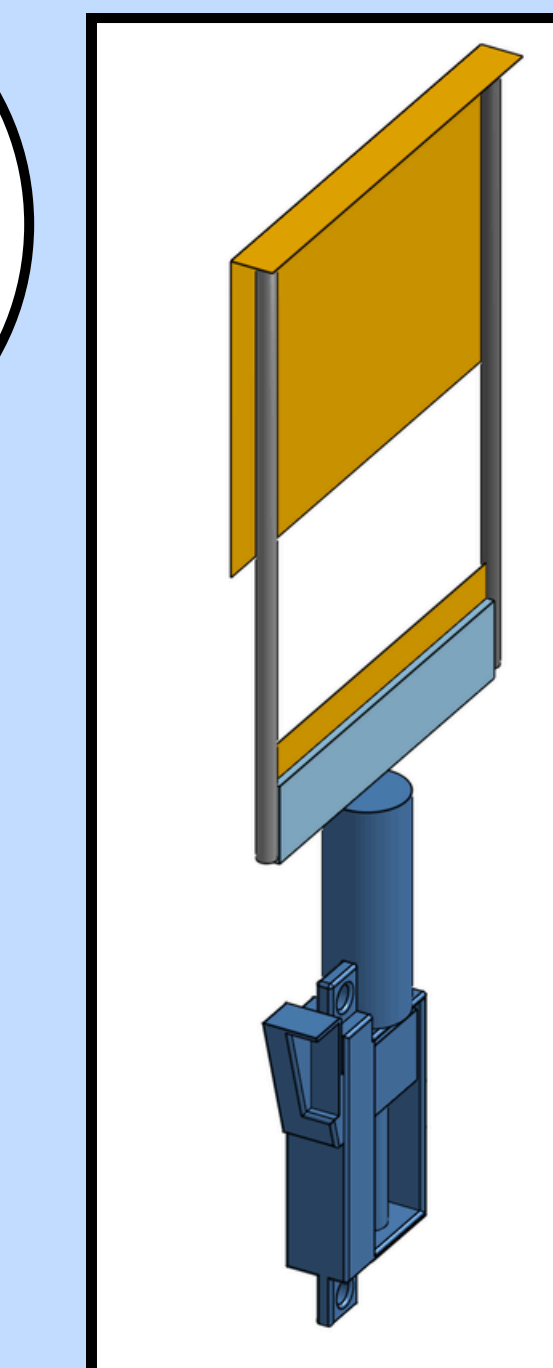
3



Buttoning

An exterior piston inserts the bracelet onto the button. A servo bends the tab into place, followed by a smaller piston that secures the button.

4



Cutting

After application, a razor blade trims the excess bracelet, allowing the user to safely remove their hand.

5



Child Engagement

Uses video, colors, and light feedback to guide the child and create a calm, engaging experience.

Transform wristband application from a manual, repetitive task into a faster, more efficient, and user-friendly process within a fast-paced daycare environment.

- Redesign the wristband process to be:
 - Faster
 - Less demanding for staff
- Reduce manual, one-by-one application
- Allow staff to focus on what matters most: supervising children

Transform the wristband application process from a manual staff task into a faster, less demanding, and kid-friendly process.

Transform wristband application from:
• Manual and repetitive

- Into:
- Faster
 - More efficient
 - Kid-friendly

Improve wristband application to increase efficiency, reduce staff burden, and enhance child experience